AWS CLOUDFORMATION LAB

Broken Web App Challenge

Resolution Documentation

Prepared by	Rutuja Naik
LinkedIn	https://www.linkedin.com/in/rutuja-naik07/
Region	US-EAST-1 (N. Virginia)
Deployment	AWS CloudFormation
Symptom	504 Gateway Time-out on the Application Load Balancer
Final result	Application loads; targets Healthy

1. Overview

This documentation outlines the step-by-step resolution process for the Broken Web App Challenge, deployed via AWS CloudFormation. The goal of the lab was to identify and fix the configuration issues preventing an application behind an Application Load Balancer (ALB) from loading properly, which resulted in a 504 Gateway Time-out error.

2. Step 1: Deploying the CloudFormation Stack

1. **Access the AWS CloudFormation console:** Navigated to the CloudFormation console in the US-EAST-1 (N. Virginia) region.
2. **Create stack:** Selected Create stack, then With new resources (standard).
3. **Template S3 URL:** Provided the CloudFormation template URL:
   `https://cloud-mastery-bootcamp.s3.amazonaws.com/lab-guides/solutions-architecture/broken-web-app-challenge.json`
4. **Stack configuration:** Named the stack using the first name (Rutuja) and submitted the creation until the status showed `CREATE_COMPLETE`.

3. Step 2: Resolving the Four Configuration Issues

Four separate misconfigurations were found across the target group, the load balancer, and the security groups. They are listed below in the order they were fixed.

Issue 1: Target Group Health Check Path Mismatch / Typo

Problem: The health check path was misconfigured (a typo), so the ALB could not get a valid response and marked the EC2 instances as Unhealthy. The instances also needed to be added as targets.

Resolution:

1. Navigated to EC2, then Target Groups, and opened the target group.
2. Opened the **Health checks** tab and clicked **Edit**.
3. Changed the health check path to `/index.html` and saved.
4. Opened the **Targets** tab, selected **Register targets**, chose the available EC2 instances, and added them as **pending** targets below, then saved.

Issue 2: Application Load Balancer: Security Group and Subnet Settings

Problem: The ALB security group did not allow the required inbound traffic, and the ALB was attached to a private subnet, so it was not reachable from the internet.

Resolution:

1. Navigated to EC2, then Load Balancers, and opened the ALB.
2. **Security group:** checked the attached security group and edited the Inbound rules to add **SSH** and **HTTP**.
3. **Subnets:** went to the **Network mapping** (Networking) section and clicked **Edit subnets**.
4. Changed the subnet selection from the **private subnet** to a **public subnet** and saved the changes.

Issue 3: Security Group Port 80 Rule (Source Restriction)

Problem: The existing port 80 rule did not use the correct source, so traffic coming from the load balancer was not being accepted.

Resolution:

1. Navigated to EC2, then Security Groups, and opened the security group.
2. Deleted the existing **port 80** inbound rule.
3. Added a new **HTTP (port 80)** rule with Source type **Custom**.
4. Set the source to the load balancer's security group by searching for the Rutuja ALB security group (the ID starting with `sg-`) and saved the rule.

Issue 4: EC2 Instance Security Group Ingress Rules

Problem: The EC2 backend instances did not have the required inbound ports open, which caused connection timeouts (504 Gateway Time-out).

Resolution:

1. Navigated to EC2, then Instances, and selected the instance.
2. Opened the **Security** tab and went to the attached security group to edit its Inbound rules.
3. As before, added the **HTTP (port 80)** and **SSH (port 22)** rules and saved.

4. Verification and Final Status

- Verified that the target status in the target group turned **Healthy**.
- Opened a new **Incognito** tab, pasted the ALB DNS name, and removed the "s" from `https` so the URL starts with `http://` (the ALB listener serves HTTP on port 80).
- The application loaded without errors and displayed the success page, served from Availability Zone `us-east-1a`.

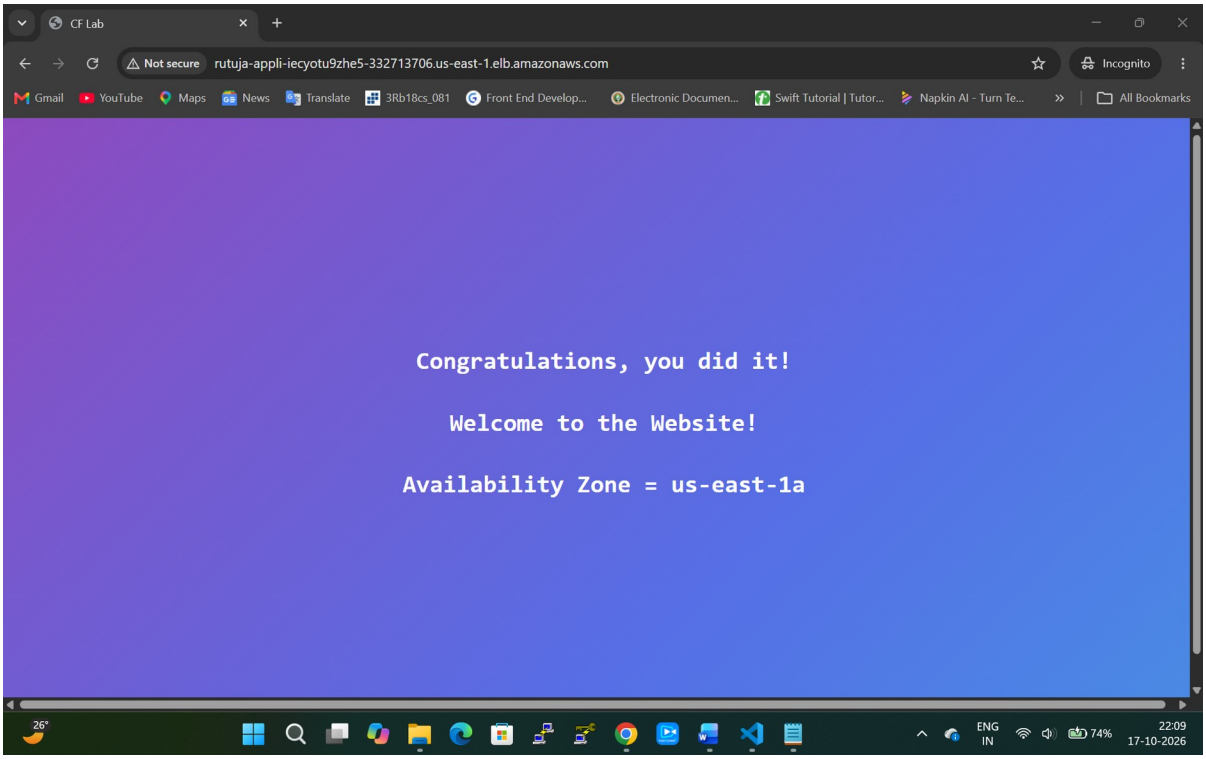


Figure 1: Application loading successfully through the ALB DNS name.

Resource Details

ALB DNS name	`rutuja-appli-iecyotu9zhe5-332713706.us-east-1.elb.amazonaws.com`
Target group	`Rutuja-ALBTa-9ZRFEFFWFKXZ`
Availability Zone	us-east-1a

Result: All four issues were resolved. The targets are Healthy and the application is reachable through the load balancer.